DP Practice Quiz M4

Question 1

You just purchased a European Put option. The strike price of the option is $15 and there is one year left until maturity. The current price of the underlying stock price is $12.5. The volatility for this stock is 45% annually, while the risk-free rate is at 1%. Calculate the sensitivity of the option price to the underlying stock price under the Black-Scholes framework. Feel free to use the provided Python code for this.

- 0.777
- -0.40
- 0.4372
- -0.56

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Question 2

Consider the following GBM process given by the following SDE:

$$dX_t = X_t(\mu dt + \sigma dW_t).$$

Suppose $Y_t = logX_t$. Applying the "Itô rule" to this process (Y_t) will result in:

- $dX = \frac{1}{Y}dY - \frac{1}{2Y^2}d[Y,Y]$
- $dY = (\mu + \frac{1}{2}\sigma^2)dt + \sigma^2 dW_t$
- $dY = \frac{1}{X}dX - \frac{1}{2X^2}d[X,X]$
- $dX_t = X_0 + (\mu - \frac{1}{2}\sigma^2)t + \sigma W_t$

Question 3

Suppose $Z_{1,t}$ and $Z_{2,t}$ are two independent processes that follow normal distributions with mean 1.15 and 0.95, respectively. What will be the mean of a process such that $W_t = Z_{1,t} - Z_{2,t}$?

- 0.42
- 2.1
- -0.2
- 0.2

Question 4

Suppose r_T is the continuously compounded interest rate. Then, the expression for the discount factor at time 0, $P_0(T)$, is:

- $P_0(T) = e^{-r_T T}$
- $P_0(T) = e^{r_T T}$

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- $P_0(T) = \frac{1}{(1+r_T)^T}$
- $P_0(T) = \frac{1}{(1-r_T)^T}$

Question 5

In the Black-Scholes model, we assume that asset prices are distributed...

- No underlying distribution is assumed.
- following a Poisson distribution
- Lognormally
- Normally

Question 6

Regarding the power of Monte Carlo methods:

- Monte Carlo methods do a better job when the number of simulations is small.
- Monte Carlo methods will only converge to a solution in the Black-Scholes world.
- Monte Carlo methods only converge to solutions when there is a closed-form analytical solution.
- Monte Carlo methods are closer to analytical solutions as the number of simulations gets bigger.

Question 7

In the standard definition of GBM we have used:

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$dS = S(\mu dt + \sigma dW_t)$, where $dW_t = \epsilon\sqrt{dt}$

What distribution does dW_t follow?

- ◯ $N(0,1)$
- ◯ $N(0,\epsilon)$
- ◯ Poisson distribution
- ◯ $N(0,dt)$

Question 8

(**Note:** Use Python to solve this question.) Consider the following Vasicek (1977) model:

$$dr = \kappa(\theta - r)dt + \sigma dz,$$

with

$$dz = \sqrt{dt}z$$

and

$$Z \sim N(0,1)$$

Use Monte Carlo to simulate one path of the interest rate in 1 year using 150 steps and a random seed equal to 42. Also use the following parameters:

$\kappa = 0.25$

$\theta = 0.02$

$\sigma = 1.80$

$r_0 = 1.875$

What is the interest rate in one year?

- ◯ 1.9482%
- ◯ 0.3031%
- ◯ 1.9304%
- ◯ 0.3332%

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Question 9

The sensitivity of options prices to the passage of time is called...

- Delta
- Theta
- There is no reason for defining such sensitivity.
- Rho

Question 10

(**Note:** Use Python to solve this question). Price a European Call option under Black-Scholes GBM dynamics. The strike price of this option is $12, with current stock price at $18.50. The risk-free rate is at 1.5% and the option will expire in 3 months. Volatility is 25%. Make sure you use a random seed equal to 0. Do this under the Black-Scholes analytical price equation and under a Monte Carlo method with 5,000 simulations. What is the difference in prices?

- Less than $0.05
- Monte Carlo estimates are higher by more than $2.
- Monte Carlo estimates are lower by more than $2.
- Monte Carlo estimates yield, as always, a lower price for the option.

Question 11

An important property of Wiener processes is...

- Wiener processes are not a special type of Markov process.

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- If X follows a Wiener process, dX_1 and dX_2, the change in X for two short time intervals depends strongly on each other.
- If X follows a Wiener process, dX, the change in X for a small period follows a Normal distribution.
- If X follows a Wiener process, dX, the change in X for a small period does not follow a Normal distribution.

Question 12

Assume a model with a general SDE of the following form:

$$dr = m(r)dt + s(r)dz$$

When is this a Vasicek (1977) model?

- $m(r) = \mu r; \quad s(r) = \sigma r$
- $m(r) = \kappa(\theta - r), \quad s(r) = \sigma\sqrt{r}$
- $m(r) = \kappa(\theta - r), \quad s(r) = \sigma$
- That is a classic GBM equation, used to model stock prices and not interest rates like Vasicek (1977) does.

Question 13

Under Black-Scholes, price a European Call option, currently ATM with three months maturity. The current price of the underlying stock price is $124. The volatility for this stock is 25% annually, while the risk-free rate is at 1.75%. Feel free to use the provided Python code for this.

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- $65.89
- $34.41
- $6.44
- $13.33

Question 14

Regarding the option prices obtained via Monte Carlo methods and a Black-Scholes closed-form solution:

- Monte Carlo estimates will generally overestimate the Black-Scholes analytical solution.
- Monte Carlo estimates will converge to an analytical solution as the number of simulations increases.
- Monte Carlo estimates will generally underestimate the Black-Scholes analytical solution.
- In case of big discrepancies between the prices from a Monte Carlo estimation and Black-Scholes analytical solution, we will always prefer to rely on the former.

Question 15

Given the following GBM:

$$dS/S = 0.1dt + 0.2dW_t$$

what are the return (R) and volatility of this stock?

- R = 20%; σ = 10%
- We cannot tell because we need some extra information on the time period.
- R = 10%; σ = 20%

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- R = 0.1; σ = 0.04

Question 16

(**Note:** Use Python to solve this question.) Consider the following Vasicek (1977) model:

$$dr = \kappa(\theta - r)dt + \sigma dz$$

with

$$dz = \sqrt{dt}z \quad \text{and} \quad Z \sim N(0,1)$$

Use Monte Carlo to simulate 25 paths of the interest rate in one year using 150 steps and a random seed equal to 42. Also use the following parameters:

$\kappa = 0.2$

$\theta = 0.025$

$\sigma = 2.75$

$r_0 = 2.85$

What is, among the 25 paths simulated, the average interest rate in 1 year?

- 2.8329%
- The values range from a minimum of -0.3401% to a maximum of 7.9631%.
- 0.2700%
- 3.228%

Question 17

In the Black-Scholes model, the closed-form expression for the sensitivity of put option prices to the underlying stock price is...

- $\mathcal{N}(d_1) - 1$
- $d_1 - \sigma\sqrt{T}$
- $\mathcal{N}(d_1)$

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- $\frac{\mathcal{N}(d_1)}{S\sigma\sqrt{T-t}}$

Question 18

(**Note:** Use Python to solve this question.) Price a European call option using a Monte Carlo method under Black-Scholes GBM dynamics. The strike price of this option is $23, with the current stock price at $22.75. The risk-free rate is at 2%, and the option will expire in seven months. Volatility is 45%. Make sure you use a random seed equal to 42. What is the price of the option after 2,500 simulations?

- $3.186
- $3.169
- $2.957
- $2.930

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Question 19

The current price of a stock is $S_0 = 100$. The annual return and volatility for this stock has historically been 10% and 20%, respectively. What will be the expected change in the price in one year if we model it with one simulation of the standard GBM?

- Use Python to solve this. You should set the random seed equal to 1 when generating random numbers.

- $1.09
- $52.48
- $42.48
- $2.07

Question 20

The Vasicek (1977) model...

- is appropriate to model interest rates.
- assumes interest rates follow a log-normal process.
- is not appropriate to model interest rates.
- cannot be used to simulate the behavior of interest rates employing Monte-Carlo methods.

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